Gradient Descent and Optimization

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Gradient Descent & Optimization Practice Exercises**

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

The quadratic loss function is defined as $L(y, \hat{y}) = (y - \hat{y})^2$. Given the true value $y = 3$ and the predicted value $\hat{y} = 5$, we substitute these values into the loss function: $L(3, 5) = (3 - 5)^2$ $L(3, 5) = (-2)^2$ $L(3, 5) = 4$ Therefore, the value of the loss function is 4.

2 **[Foundational] • Target: Loss Function Derivatives and Stationary Points**

Consider a discrete random variable X with the following probability mass function (PMF): $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. What is the value of $P(X \leq 2)$?

The probability mass function is given by $P(X = x) = \frac{x}{10}$ for $x \in \{1, 2, 3, 4\}$. We need to find $P(X \leq 2)$. This is the sum of probabilities for $X = 1$ and $X = 2$. $P(X \leq 2) = P(X = 1) + P(X = 2)$. $P(X = 1) = \frac{1}{10}$ and $P(X = 2) = \frac{2}{10}$. Therefore, $P(X \leq 2) = \frac{1}{10} + \frac{2}{10} = \frac{3}{10}$.

3 **[Foundational] • Target: Multivariate Gradients and Directional Derivatives**

The sigmoid activation function is $\sigma(z) = \frac{1}{1 + e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 1.20$: $\sigma'(z_0) = (1.20)(1 - 1.20) = (1.20)(-0.20) = -0.2400$. Final Answer: -0.2400

4 **[Foundational] • Target: Gradient Descent Step Updates**

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

Odds are $\frac{p}{1-p} = \frac{0.30}{0.70} = 0.4286$. Taking natural logarithm: $\ln(0.4286) = -0.847$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

5 **[Foundational] • Target: Momentum and Advanced Optimizers**

For a bivariate feature dataset, $\text{Cov}(x, y) = 36$, $\text{Var}(x) = 12$, $\bar{x} = 5$, and $\bar{y} = 35$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \frac{36}{12} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 35 - (3)(5) = 35 - 15 = 20$. Final Answer: $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 20$

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6 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In training a language model token predictor, a binary token occurs $k = 15$ times across $n = 50$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

For a Bernoulli likelihood $L(p) = p^k(1-p)^{n-k}$, the log-likelihood is $\ln L = k\ln p + (n-k)\ln(1-p)$. Setting derivative to zero: $\frac{k}{p} - \frac{n-k}{1-p} = 0 \Rightarrow \hat{p} = \frac{k}{n} = \frac{15}{50} = 0.300$.

7 [Foundational] • Target: Loss Function Derivatives and Stationary Points

A spam classification model identifies $TP = 120$ genuine spam emails and $FP = 40$ legitimate emails falsely flagged as spam. What is the precision of this model?

Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP+FP} = \frac{120}{120+40} = \frac{120}{160} = 75.0\%$. Final Answer: 75.0%

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

By the multiplication rule of conditional probability: $P(\text{Defect} \cap \text{Positive}) = P(\text{Defect}) \times P(\text{Positive}|\text{Defect}) = 0.25 \times 0.90 = 0.2250$.

9 [Intermediate] • Target: Gradient Descent Step Updates

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

By Bayes' Rule: $P(A|+) = \frac{P(+|A)P(A)}{P(+|A)P(A) + P(+|egA)P(egA)} = \frac{0.95 \times 0.02}{0.95(0.02) + 0.05(0.98)} = \frac{0.0190}{0.0680} = 0.279$. Final Answer: 0.279

10 [Intermediate] • Target: Momentum and Advanced Optimizers

Compute the Manhattan (L_1) norm and Euclidean (L_2) norm of the word vector $\mathbf{w} = [10, -15]^T$.

L_1 norm is the sum of absolute values: $|10| + |-15| = 25$. L_2 norm is the square root of sum of squares: $\sqrt{(10)^2 + (-15)^2} = \sqrt{100 + 225} = \sqrt{325}$.

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11 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

Consider a simple quadratic loss function $L(w) = (w^2 - 4w + 5)$. We want to find the minimum of this loss function using gradient descent. The gradient of the loss function with respect to w is given by $\frac{dL}{dw} = 2w - 4$. If we start at an initial weight $w_0 = 5$ and use a learning rate $\alpha = 0.5$, what will be the value of the weight w_1 after the first update step, i.e., $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$? Show your complete algebraic work.

Final Answer: 4. The loss function is $L(w) = w^2 - 4w + 5$. The gradient is $\frac{dL}{dw} = 2w - 4$. We are given an initial weight $w_0 = 5$ and a learning rate $\alpha = 0.5$. The gradient at w_0 is $\frac{dL}{dw}(w_0) = 2(5) - 4 = 10 - 4 = 6$. The update rule for gradient descent is $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$. Substituting the values, we get $w_1 = 5 - 0.5 \times 6 = 5 - 3 = 2$.

12 **[Intermediate]** • **Target: Loss Function Derivatives and Stationary Points**

Consider a univariate regression model where the predicted output $\hat{y}$ is given by $\hat{y} = ax + b$. The loss function is the Mean Squared Error (MSE), defined as $L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. $*Foraasingledatapoint*(x_1, y_1) = (2, 5), *thelossis*L = (5 - (2a + b))^2. *Thegradientofthelossfunctionwithrespecttoais*\frac{\partial L}{\partial a} = -2(5 - (2a + b)) \cdot 2. *Ifduringgradientdescent*, *withalearningrateof*\eta = 0.1, *theparameters*a *and* b *areupdatedfrom*a_{old} = 1 *and* b_{old} = 1, *to* a_{new} = 0.8 *and* b_{new} = 0.8, *whatisthevalueof*b *thatwouldhaveresultedin*\frac{\partial L}{\partial b} = 0 *forthissingledatapoint*, *assuming*a=1 *and* b=1 *werethevalues* * *before* * *thegradientcalculationthatledtotheupdateto* a_{new}=0.8, b_{new}=0.8$? Show your complete algebraic work.

Final Answer: 5. The loss function for a single data point is $L = (y_1 - (ax_1 + b))^2$. $*Given*(x_1, y_1) = (2, 5), *so*L = (5 - (2a + b))^2. *Thegradientofthelosswithrespecttob* *is*\frac{\partial L}{\partial b} = 2(5 - (2a + b)) \cdot (-1) = -2(5 - 2a - b). *Weareaskedtofindthevalueof*b *forwhich*\frac{\partial L}{\partial b} = 0, *assuming*a=1 *and* b=1 *weretheparameters* * *before* * *theupdatestepthatledto* a_{new}=0.8, b_{new}=0.8. *Thismeansweevaluatethegradientat*a=1$ and the *unknown.

13 **[Intermediate]** • **Target: Multivariate Gradients and Directional Derivatives**

Consider the scalar loss function $L(w) = 6w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 10$.

Final Answer: $ablaL(10) = 112$. Differentiate with respect to w: $\frac{dL}{dw} = 2(6)w - 8 = 12w - 8$. At $w = 10$: $ablaL = 12(10) - 8 = 120 - 8 = 112$.

14 **[Intermediate]** • **Target: Gradient Descent Step Updates**

Consider the scalar loss function $L(w) = 7w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 12$.

Final Answer: $ablaL(12) = 160$. Differentiate with respect to w: $\frac{dL}{dw} = 2(7)w - 8 = 14w - 8$. At $w = 12$: $ablaL = 14(12) - 8 = 168 - 8 = 160$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Momentum and Advanced Optimizers** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

Consider the scalar loss function $L(w) = 5w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 8$.

Final Answer: $ablaL(8) = 72$. Differentiate with respect to w: $\frac{dL}{dw} = 2(5)w - 8 = 10w - 8$. At $w = 8$: $ablaL = 10(8) - 8 = 80 - 8 = 72$.

16 **[Challenging]** • **Target: Gradient Descent & Optimization Practice Exercises** • **Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

Consider the scalar loss function $L(w) = 3w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 4$.

Final Answer: $ablaL(4) = 16$. Differentiate with respect to w: $\frac{dL}{dw} = 2(3)w - 8 = 6w - 8$. At $w = 4$: $ablaL = 6(4) - 8 = 24 - 8 = 16$.

17 **[Challenging]** • **Target: Loss Function Derivatives and Stationary Points** • **Cross-Lesson: 6.01 Linear Regression and OLS**

Consider the scalar loss function $L(w) = 4w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 6$.

Final Answer: $ablaL(6) = 40$. Differentiate with respect to w: $\frac{dL}{dw} = 2(4)w - 8 = 8w - 8$. At $w = 6$: $ablaL = 8(6) - 8 = 48 - 8 = 40$.

18 **[Challenging]** • **Target: Multivariate Gradients and Directional Derivatives** • **Cross-Lesson: 7.01 Logistic Regression and Classification**

Consider the scalar loss function $L(w) = 2w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 2$.

Final Answer: $ablaL(2) = 0$. Differentiate with respect to w: $\frac{dL}{dw} = 2(2)w - 8 = 4w - 8$. At $w = 2$: $ablaL = 4(2) - 8 = 8 - 8 = 0$.

19 **[Challenging]** • **Target: Gradient Descent Step Updates** • **Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.

Final Answer: $w = 2$. The given loss function is a quadratic function $L(w) = w^2 - 4w + 3$. This represents a parabola opening upwards. The minimum value of a quadratic function of the form $aw^2 + bw + c$ occurs at $w = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -4$. Therefore, the minimum occurs at $w = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Alternatively, we can find the minimum by taking the derivative and setting it to zero: $L'(w) = 2w - 4$. Setting $L'(w) = 0$ gives $2w - 4 = 0$, which implies $2w = 4$.

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20 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function? Show your complete algebraic work.

Final Answer: 2. To find the minimum of the quadratic loss function $L(w) = 3w^2 - 12w + 10$, we can find the derivative with respect to w and set it to zero. The derivative of $L(w)$ with respect to w is: $\frac{dL}{dw} = 6w - 12$. Setting the derivative to zero to find the critical point: $6w - 12 = 0$ $6w = 12$ $w = \frac{12}{6}$ $w = 2$. The second derivative is $\frac{d^2L}{dw^2} = 6$, which is positive, confirming that $w = 2$ corresponds to a minimum.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value \(w = 10 \). If the gradient of the loss is \(\frac{\partial L}{\partial w} = 4 \) and the learning rate is \(\eta = 0.1 \), find the updated weight \(w' \).

a. \(10.40 \)

b. **\(9.60 \)** ✓

c. \(6.00 \)

d. \(9.90 \)

Using the gradient descent update rule: \(w' = w - \eta \frac{\partial L}{\partial w} = 10 - (0.1)(4) = 10 - 0.40 = 9.60 \).

22 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value \(w = 20 \). If the gradient of the loss is \(\frac{\partial L}{\partial w} = 8 \) and the learning rate is \(\eta = 0.2 \), find the updated weight \(w' \).

a. \(12.00 \)

b. \(21.60 \)

c. **\(18.40 \)** ✓

d. \(19.80 \)

Using the gradient descent update rule: \(w' = w - \eta \frac{\partial L}{\partial w} = 20 - (0.2)(8) = 20 - 1.60 = 18.40 \).

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23 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 30$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 12$ and the learning rate is $\eta = 0.3$, find the updated weight w'.

a. 18.00 **b.** 33.60

c. **26.40** ✓ **d.** 29.70

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 30 - (0.3)(12) = 30 - 3.60 = 26.40$.

24 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 40$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 16$ and the learning rate is $\eta = 0.4$, find the updated weight w'.

a. 24.00 **b.** 46.40

c. **33.60** ✓ **d.** 39.60

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 40 - (0.4)(16) = 40 - 6.40 = 33.60$.

25 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 50$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 20$ and the learning rate is $\eta = 0.5$, find the updated weight w'.

a. 30.00 **b.** 60.00

c. **40.00** ✓ **d.** 49.50

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 50 - (0.5)(20) = 50 - 10.00 = 40.00$.

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26 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value $w = 60$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 24$ and the learning rate is $\eta = 0.6$, find the updated weight w'.

a. 36.00

b. 74.40

c. **45.60** ✓

d. 59.40

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 60 - (0.6)(24) = 60 - 14.40 = 45.60$.

27 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the scalar loss function $L(w) = 2w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 2$.

a. $\nabla L(2) = 16$

b. $\nabla L(2) = 8$

c. **$\nabla L(2) = 0$** ✓

d. $\nabla L(2) = -4$

Differentiate with respect to w: $\frac{dL}{dw} = 2(2)w - 8 = 4w - 8$. At $w = 2$: $\nabla L = 4(2) - 8 = 8 - 8 = 0$.

28 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the scalar loss function $L(w) = 3w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 4$.

a. **$\nabla L(4) = 16$** ✓

b. $\nabla L(4) = 24$

c. $\nabla L(4) = 32$

d. $\nabla L(4) = 4$

Differentiate with respect to w: $\frac{dL}{dw} = 2(3)w - 8 = 6w - 8$. At $w = 4$: $\nabla L = 6(4) - 8 = 24 - 8 = 16$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Loss Function Derivatives and Stationary Points

Find the stationary point (global minimum) of the scalar mean squared error loss $L(w) = 3(w-4)^2 + 7$.

a. $w = 4$ ✓

b. $w = 7$

c. $w = 3$

d. $w = 0$

$L'(w) = 6(w-4) = 0 \Rightarrow w = 4$. $L''(4) = 6 > 0$, so $w = 4$ is the minimum.

30 [Intermediate] • Target: Loss Function Derivatives and Stationary Points

For the loss function $L(w) = w^4 - 4w^2 + 6$, find all values of w where the gradient is zero.

a. $w = 0, \pm\sqrt{2}$ ✓

b. $w = \pm 2$

c. $w = 0, \pm 2$

d. $w = \pm 1$

$L'(w) = 4w^3 - 8w = 4w(w^2 - 2) = 0 \Rightarrow w = 0$ or $w = \pm\sqrt{2}$.

31 [Intermediate] • Target: Loss Function Derivatives and Stationary Points

In optimization, a critical point has $L'(w) = 0$ and $L''(w) = -8$. What type of point is this?

a. **Local Maximum** ✓

b. Local Minimum

c. Saddle Point

d. Inflection Point

A negative second derivative implies concavity downwards, confirming a local maximum.

32 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

Compute the gradient vector ∇L of $L(w_1, w_2) = 3w_1^2 + 5w_2^2 - 2w_1w_2$ evaluated at $(w_1, w_2) = (2, 1)$.

a. $[10, 6]^T$ ✓

b. $[12, 10]^T$

c. $[8, 8]^T$

d. $[14, 4]^T$

$\frac{\partial L}{\partial w_1} = 6w_1 - 2w_2 = 12 - 2 = 10$. $\frac{\partial L}{\partial w_2} = 10w_2 - 2w_1 = 10 - 4 = 6$.

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33 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

In which direction does the negative gradient vector $-\nabla L(\mathbf{w})$ point?

a. The direction of steepest decrease of the loss function ✓

b. The direction of steepest increase of the loss function

c. Orthogonal to the level curves in the direction of highest loss

d. Along the tangent line of the contour

∇L points in the direction of greatest increase; its negation $-\nabla L$ points in the direction of steepest descent.

34 [Intermediate] • Target: Multivariate Gradients and Directional Derivatives

Compute the directional derivative of $f(x, y) = x^2 y$ at $(1, 2)$ in the unit direction $\mathbf{u} = [0.8, 0.6]^T$.

a. 3.8 ✓

b. 4.2

c. 2.6

d. 5.0

$\nabla f = [2xy, x^2]^T = [4, 1]^T$. Directional derivative $= \nabla f \cdot \mathbf{u} = 4(0.8) + 1(0.6) = 3.2 + 0.6 = 3.8$.

35 [Intermediate] • Target: Gradient Descent Step Updates

A model weight has value $w_0 = 5.0$. If $\frac{\partial L}{\partial w} = 4.0$ and learning rate is $\eta = 0.1$, find the updated weight w_1.

a. 4.6 ✓

b. 5.4

c. 4.0

d. 4.9

Update: $w_1 = w_0 - \eta \nabla L = 5.0 - 0.1(4.0) = 5.0 - 0.4 = 4.6$.

36 [Intermediate] • Target: Gradient Descent Step Updates

Consider the quadratic loss $L(w) = \frac{1}{2} a w^2$ with $a = 10$. What is the maximum learning rate η before gradient descent diverges?

a. $\eta < 0.2$ ✓

b. $\eta < 0.1$

c. $\eta < 0.05$

d. $\eta < 1.0$

Update is $w_{t+1} = w_t(1 - \eta a)$. Stability requires $|1 - \eta a| < 1 \Rightarrow -1 < 1 - 10\eta < 1 \Rightarrow 0 < \eta < 2/10 = 0.2$.

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37 [Intermediate] • Target: Gradient Descent Step Updates

Perform one vector gradient descent update from $\mathbf{w}_0 = [1, 2]^T$ with $\nabla L = [2, -4]^T$ and $\eta = 0.25$.

a. $[0.5, 3.0]^T$ ✓

b. $[1.5, 1.0]^T$

c. $[0.5, 1.0]^T$

d. $[1.0, 3.0]^T$

$\mathbf{w}_1 = [1, 2]^T - 0.25[2, -4]^T = [1 - 0.5, 2 - (-1)]^T = [0.5, 3.0]^T$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 7.01 Logistic Regression and Classification

In Polyak momentum, the velocity update is $\mathbf{v}_{t+1} = \beta \mathbf{v}_t + \eta \nabla L$. If $\beta = 0.9$, $\mathbf{v}_0 = 2.0$, $\eta = 0.1$, and $\nabla L = 4.0$, find $\mathbf{v}_1$.

a. 2.2 ✓

b. 2.0

c. 1.8

d. 2.6

$\mathbf{v}_1 = 0.9(2.0) + 0.1(4.0) = 1.8 + 0.4 = 2.2$.

39 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Why does the Adam optimizer maintain running exponential averages of both first moments (mean) and second raw moments (uncentered variance)?

a. **To adapt per-parameter learning rates inversely proportional to gradient magnitude** ✓

b. To guarantee convergence to the global minimum in non-convex losses

c. To replace backpropagation with finite differences

d. To eliminate the need for mini-batch sampling

The second moment scales steps down for parameters with large, frequent gradients and up for sparse gradients.

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40 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 1.01 Vectors and Embeddings in AI

What is the condition on the Hessian matrix $\mathbf{H} = \nabla^2 L(\mathbf{w})$ for a loss function to be strictly convex everywhere?

a. $\mathbf{H}$ is positive definite everywhere (all eigenvalues > 0) ✓

b. $\mathbf{H}$ has determinant equal to zero

c. $\mathbf{H}$ has trace equal to 1

d. $\mathbf{H}$ is skew-symmetric

Strict convexity requires a positive definite Hessian: $\mathbf{v}^T\mathbf{H}\mathbf{v} > 0$ for all non-zero $\mathbf{v}$.

41 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

A 2D loss surface is defined as $L(w_1, w_2) = 6w_1^2 + 11w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [26, -24]^T$

b. $ablaL = [24, 22]^T$

c. $ablaL = [12, -11]^T$

d. $ablaL = [24, -22]^T$ ✓

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(6)w_1 = 12(2) = 24$; $\frac{\partial L}{\partial w_2} = 2(11)w_2 = 22(-1) = -22$. Therefore, $ablaL = [24, -22]^T$.

42 [Challenging] • Target: Loss Function Derivatives and Stationary Points • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A 2D loss surface is defined as $L(w_1, w_2) = 7w_1^2 + 13w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [30, -28]^T$

b. $ablaL = [28, 26]^T$

c. $ablaL = [14, -13]^T$

d. $ablaL = [28, -26]^T$ ✓

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(7)w_1 = 14(2) = 28$; $\frac{\partial L}{\partial w_2} = 2(13)w_2 = 26(-1) = -26$. Therefore, $ablaL = [28, -26]^T$.

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43 [Challenging] • Target: Multivariate Gradients and Directional Derivatives • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A 2D loss surface is defined as $L(w_1, w_2) = 5w_1^2 + 9w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [22, -20]^T$

b. $ablaL = [20, 18]^T$

c. $ablaL = [10, -9]^T$

d. $ablaL = [20, -18]^T$ ✓

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(5)w_1 = 10(2) = 20$; $\frac{\partial L}{\partial w_2} = 2(9)w_2 = 18(-1) = -18$. Therefore, $ablaL = [20, -18]^T$.

44 [Challenging] • Target: Gradient Descent Step Updates • Cross-Lesson: 6.01 Linear Regression and OLS

A 2D loss surface is defined as $L(w_1, w_2) = 3w_1^2 + 5w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [14, -12]^T$

b. $ablaL = [12, 10]^T$

c. $ablaL = [6, -5]^T$

d. $ablaL = [12, -10]^T$ ✓

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(3)w_1 = 6(2) = 12$; $\frac{\partial L}{\partial w_2} = 2(5)w_2 = 10(-1) = -10$. Therefore, $ablaL = [12, -10]^T$.

45 [Challenging] • Target: Momentum and Advanced Optimizers • Cross-Lesson: 7.01 Logistic Regression and Classification

A 2D loss surface is defined as $L(w_1, w_2) = 4w_1^2 + 7w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.

a. $ablaL = [18, -16]^T$

b. $ablaL = [16, 14]^T$

c. $ablaL = [8, -7]^T$

d. $ablaL = [16, -14]^T$ ✓

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(4)w_1 = 8(2) = 16$; $\frac{\partial L}{\partial w_2} = 2(7)w_2 = 14(-1) = -14$. Therefore, $ablaL = [16, -14]^T$.

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